Polarization & Magnetization

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The Equations

All the equations we will discuss in this note:

$$\begin{cases} \varrho_b = -\boldsymbol{\nabla}\cdot\mathbf{P} \\ \sigma_b = \mathbf{P}\cdot\hat{\mathbf{n}} \end{cases} \qquad \begin{cases} \mathbf{J}_b = \boldsymbol{\nabla}\times\mathbf{M} \\ \mathbf{K}_b = \mathbf{M}\times\hat{\mathbf{n}} \end{cases} \qquad \begin{cases} \mathbf{D} = \epsilon_0\mathbf{E}+\mathbf{P} \\ \mathbf{H} = \dfrac{1}{\mu_0}\mathbf{B}-\mathbf{M} \end{cases} \qquad \begin{cases} \mathbf{D} = \epsilon_0(1+\chi_e)\mathbf{E} \\ \mathbf{B} = \mu_0(1+\chi_m)\mathbf{H} \end{cases}$$

1 Polarization of the Dielectric

The potential of the dipole system in an electric field is written as:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\mathbf{P}(\mathbf{r}')\cdot\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}d\tau'. \tag{1}$$

We can apply the Delta Functions Identities (iii) on the integral, then it becomes:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\mathbf{P}(\mathbf{r}')\cdot\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}d\tau' = \frac{1}{4\pi\epsilon_0}\int \mathbf{P}(\mathbf{r}')\cdot \underbrace{\boldsymbol{\nabla}'\left(\frac{1}{\mathscr{r}}\right)}_{\boldsymbol{\nabla}'\left(\frac{1}{\mathscr{r}}\right)=\frac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}} d\tau'.$$

Note that the analysis inside the integral is carried out in the *source coordinates*. This is because the polarization phenomenon occurs in the source region, and our focus is on that region. Therefore, the gradient operator must be written with respect to the source coordinates, and

$$\boldsymbol{\nabla} \;\rightarrow\; \boldsymbol{\nabla}'$$

is used to indicate differentiation with respect to the source variables. By integration by parts, the potential becomes

$$V = \frac{1}{4\pi\epsilon_0}\int \mathbf{P}(\mathbf{r}')\cdot\boldsymbol{\nabla}'\left(\frac{1}{\mathscr{r}}\right) = \frac{1}{4\pi\epsilon_0}\left[\int_{\mathcal{V}}\boldsymbol{\nabla}'\cdot\left(\frac{\mathbf{P}(\mathbf{r}')}{\mathscr{r}}\right)d\tau' - \int_{\mathcal{V}}\frac{1}{\mathscr{r}}\boldsymbol{\nabla}\cdot\mathbf{P}(\mathbf{r}')d\tau'\right]. \tag{2}$$

We can rewrite the first integral in (2), that is:

$$\int_{\mathcal{V}}\boldsymbol{\nabla}'\cdot\left(\frac{\mathbf{P}(\mathbf{r}')}{\mathscr{r}}\right)d\tau' = \oint_{\partial\mathcal{V}}\frac{\mathbf{P}(\mathbf{r})}{\mathscr{r}}\cdot d\mathbf{a} = \oint_{\partial\mathcal{V}}\frac{\mathbf{P}(\mathbf{r})\cdot\hat{\mathbf{n}}}{\mathscr{r}}\cdot da. \tag{3}$$

(2) and (3) indicate that

$$\boxed{\begin{cases} \sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} \\ \\ \varrho_b = -\boldsymbol{\nabla} \cdot \mathbf{P}. \end{cases}} \tag{4}$$

Here, σ_b represents the bound surface charge density that arises when a material becomes polarized under the influence of an electric field. The polarization separates positive and negative charges within the material, creating an electric dipole structure. As a consequence, opposite charge densities naturally appear on the boundaries where the separation occurs.

Similarly, within a given volume, regardless of whether the charge distribution is uniform or not, we can always enclose a region and compute the total charge distribution inside it. From the perspective of polarization, the divergence of the polarization field describes how the dipoles terminate within the enclosed region. In general, the direction of $\boldsymbol{\nabla} \cdot \mathbf{P}$ is opposite to the effective bound charge contained in the volume. For example, if the total enclosed bound charge is negative, the polarization vectors tend to point outward, indicating that positive charges appear outside the volume while negative charges accumulate inside. In this case, the divergence of $\mathbf{P}$ is positive, whereas the enclosed charge is negative.

Then we can rewrite the potential as

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \oint_{\partial\mathcal{V}} \frac{\sigma_b}{\boldsymbol{\mathscr{r}}}\, da' + \frac{1}{4\pi\epsilon_0} \int_{\mathcal{V}} \frac{\varrho_b}{\boldsymbol{\mathscr{r}}}\, d\tau' \tag{5}$$

In other words, a polarized dielectric can be treated by introducing the bound surface charge density and the bound volume charge density produced by the polarization. By accounting for these effective charge distributions, we can compute the overall electric effect of the material without having to explicitly sum the contributions from each infinitesimal dipole moment inside the dielectric.

2 Magnetization of the Matters

The vector potential of the magnetic dipole system in an magnetic field is written as:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{M}(\mathbf{r}') \times \hat{\boldsymbol{\mathscr{r}}}}{\boldsymbol{\mathscr{r}}^2}\, d\tau'. \tag{6}$$

Likewise, it can be written as

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int_{\mathcal{V}} \mathbf{M}(\mathbf{r}') \times \boldsymbol{\nabla}' \left(\frac{1}{\boldsymbol{\mathscr{r}}} \right) d\tau'.$$

By the Products Rules (v), the integral becomes:

$$\mathbf{A} = \frac{\mu_0}{4\pi}\left[\int_{\mathcal{V}} \frac{1}{\mathscr{r}} \boldsymbol{\nabla} \times \mathbf{M}(\mathbf{r}')\, d\tau' - \underbrace{\int_{\mathcal{V}} \boldsymbol{\nabla} \times \left(\frac{\mathbf{M}(\mathbf{r}')}{\mathscr{r}}\right) d\tau'}_{\text{Green's Identities and Some Integrals (iv)}}\right]$$

$$= \frac{\mu_0}{4\pi}\left[\int_{\mathcal{V}} \frac{1}{\mathscr{r}} \boldsymbol{\nabla} \times \mathbf{M}(\mathbf{r}')\, d\tau' + \oint_{\partial\mathcal{V}} \frac{\mathbf{M}(\mathbf{r}')}{\mathscr{r}} \times d\mathbf{a}'\right].$$

So, the vector potential $\mathbf{A}$ is:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int_{\mathcal{V}} \frac{1}{\mathscr{r}} \boldsymbol{\nabla} \times \mathbf{M}(\mathbf{r}')\, d\tau' + \frac{\mu_0}{4\pi}\oint_{\partial\mathcal{V}} \frac{\mathbf{M}(\mathbf{r}') \times \hat{\mathbf{n}}}{\mathscr{r}} da' \tag{7}$$

(7) indicates that

$$\boxed{\begin{cases} \mathbf{J}_b = \boldsymbol{\nabla} \times \mathbf{M} \\ \\ \mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}. \end{cases}} \tag{8}$$

The physical interpretation is the following. When a uniform material is magnetized by an external magnetic field, microscopic current loops are induced throughout the material. These loops represent the magnetic dipole moments associated with the magnetization $\mathbf{M}$. If the magnetization is uniform, the circulating currents in the interior of the material cancel each other pairwise, since neighboring loops carry currents in opposite directions along their shared boundaries. As a result, the only remaining current appears at the outer boundary of the material. This leads to an effective surface current density, and $\hat{\mathbf{n}}$ is the outward normal vector of the surface. Physically, the cross product indicates that the direction of the surface current is perpendicular to both the magnetization direction and the outward normal.

If the magnetization is non-uniform, however, the microscopic current loops in neighboring regions do not cancel perfectly. As a consequence, a net current remains within the bulk of the material. This residual current is described by the bound volume current. In this sense, the curl of the magnetization field measures the local imbalance of microscopic current loops. When $\mathbf{M}$ varies in space, the cancellation between neighboring loops becomes incomplete, producing a net circulating current density in the interior of the material. We can use some method to explain that. The net current in the $i-$th direction is[1]:

$$I_i = \underbrace{(\partial_j M_k) dx_j}_{\text{Total derivative}} dx_k - (\partial_k M_j) dx_k dx_j,$$

[1]Note that for the infinitesimal volume, the current on the surface is $M \times$length because $I \times$area $= m$ and $m/$volume $= M = I \times$ area/volume. Therefore, $I = M \times$ length.

and this leads to

$$J_i = \underbrace{\partial_j M_k - \partial_k M_j}_{\text{Different direction of current}} \qquad (i,j,k) \text{ cyclic}$$

$$= \epsilon_{ijk}\partial_j M_k = \big(\boldsymbol{\nabla}\times\mathbf{M}\big)_i$$

The expression of the vector potential is now rewritten in the form of the current densities:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int_{\mathcal{V}} \frac{\mathbf{J}_b(\mathbf{r}')}{\mathscr{r}}\, d\tau' + \frac{\mu_0}{4\pi}\oint_{\partial\mathcal{V}} \frac{\mathbf{K}_b(\mathbf{r}')}{\mathscr{r}}\, da' \tag{9}$$

3 The Rearrangement of the Fields in Matters

3.1 Electric Displacement

The discussions above show the phenomena of the matter in the electric field. The field inside the dielectric is influenced by the field outside, and it will be polarized. Therefore, the different electric fields is complicated to deal with; to make it simple, we can further rearrange some quantities. By Gauss's law, the divergence of the total electric field (also consider the field in the matter) is given by

$$\epsilon_0 \boldsymbol{\nabla}\cdot\mathbf{E} = \varrho(\mathbf{r}) = \varrho_f(\mathbf{r}) + \varrho_b(\mathbf{r}), \tag{10}$$

where ϱ_f represents the "free" charges which are not affected by the polarized matters and fields. If we only focus on the charges that don't involve slaving, then we can rewrite (10) as:

$$\epsilon_0 \boldsymbol{\nabla}\cdot\mathbf{E} = \varrho_f + \varrho_b = \varrho_f - \boldsymbol{\nabla}\cdot\mathbf{P} \Rightarrow \boldsymbol{\nabla}\cdot\big(\epsilon_0\mathbf{E} + \mathbf{P}\big) = \varrho_f. \tag{11}$$

where we define

$$\boxed{\mathbf{D} \coloneqq \epsilon_0\mathbf{E} + \mathbf{P}.} \tag{12}$$

Hence, Gauss's law is modified to the "Polarized Version":

$$\boldsymbol{\nabla}\cdot\mathbf{D} = \varrho_f, \quad \int \mathbf{D}\cdot d\mathbf{a} = Q_{f_{\text{enc}}} \tag{13}$$

If the matter is a **linear dielectric**, then the dipole moment is proportional to the electric field, which can be written as:

$$\mathbf{P}_{\text{lin}} = \epsilon_0\chi_e\mathbf{E} \tag{14}$$

where χ_e is called the **electric susceptibility**. Substituting this into (12), then the electric displacement becomes

$$\boxed{\mathbf{D} = \epsilon_0\mathbf{E} + \epsilon_0\chi_e\mathbf{E} = \epsilon_0(1+\chi_e)\mathbf{E} = \epsilon_0\epsilon_r\mathbf{E} = \epsilon\mathbf{E}.} \tag{15}$$

where we define

$$\begin{cases} \epsilon \coloneqq \epsilon_0(1+\chi_e) \\ \\ \epsilon_r \coloneqq 1+\chi_e. \end{cases} \tag{16}$$

ϵ is called **permittivity of the material**, while ϵ_r is called the relative permittivity. Note that although we can "think of" electric displacement as the electric field, but in fact, the displacement field is not always parallel to the electric field. That is,

$$\boldsymbol{\nabla} \times \mathbf{D} = \boldsymbol{\nabla} \times (\epsilon_0 \mathbf{E} + \mathbf{P}) = \boldsymbol{\nabla} \times \mathbf{P}. \tag{17}$$

In a linear dielectric, this becomes

$$\begin{aligned} \boldsymbol{\nabla} \times \mathbf{D} = \boldsymbol{\nabla} \times [\epsilon_0(1+\chi_e)\mathbf{E}] = \epsilon_0 \boldsymbol{\nabla} \times \mathbf{E} + \boldsymbol{\nabla} \times (\epsilon_0 \chi_e \mathbf{E}) \\ = \epsilon_0 \chi_e (\boldsymbol{\nabla} \times \mathbf{E}) - \epsilon_0 \mathbf{E} \times \boldsymbol{\nabla} \chi_e = -\epsilon_0 \mathbf{E} \times \boldsymbol{\nabla} \chi_e. \end{aligned} \tag{18}$$

Whether the dielectric is linear or not, the curl of the electric displacement depends on the dipole moment or the consistency of the electric susceptibility.

3.2 Auxiliary Field

Likewise, the current density established by the magnetization is called the bound current $\mathbf{J}_b$; while the current caused by other magnetic field is called $\mathbf{J}_f$, A.K.A. free current. That is:

$$\mathbf{J} = \mathbf{J}_f + \mathbf{J}_b, \quad \mu_0 \mathbf{J} = \mu_0 \mathbf{J}_f + \mu_0 \mathbf{J}_b. \tag{19}$$

By Ampère's law and (**??**), then (19) becomes:

$$\boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \mathbf{J} = \mu_0 \mathbf{J}_b + \mu_0 \boldsymbol{\nabla} \times \mathbf{M}, \tag{20}$$

therefore, the free current density can be written as:

$$\mathbf{J}_f = \frac{1}{\mu_0} \boldsymbol{\nabla} \times \mathbf{B} - \boldsymbol{\nabla} \times \mathbf{M} = \boldsymbol{\nabla} \times \left(\frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \right). \tag{21}$$

We define

$$\boxed{\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M},} \tag{22}$$

and this is called the **Auxiliary Field**. As the electric displacement, we use the auxiliary field to deal with the free current established by other magnetic field. In terms of $\mathbf{H}$, Ampère's law becomes

$$\boldsymbol{\nabla} \times \mathbf{H} = \mathbf{J}_f, \quad \oint \mathbf{H} \cdot d\boldsymbol{\ell} = I_{f_{\text{enc}}}. \tag{23}$$

In addition, when the matter in the magnetic field is a linear medium, then the magnetic dipole moment $\mathbf{M}$ can be written as:

$$\mathbf{M} = \chi_m \mathbf{H}, \tag{24}$$

where χ_m is called the **magnetic susceptibility**. By the definition of the auxiliary field $\mathbf{H}$, we have

$$\boxed{\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu_0(1 + \chi_m)\mathbf{H} = \mu \mathbf{H},} \tag{25}$$

where μ is called the **permeability of the material**. It's defined as

$$\mu \coloneqq \mu_0(1 + \chi_m) \tag{26}$$

This is a little bit different from the linear dielectric $\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$ This choice is motivated by the physical mechanisms underlying polarization and magnetization.
In an electric medium, polarization arises from the displacement of bound charges under the influence of the electric field. Since the electric field $\mathbf{E}$ directly acts on charges through the Lorentz force, it is the natural variable that determines the response of the material. Therefore, expressing the polarization as a function of $\mathbf{E}$ provides a direct and physically transparent constitutive relation.
Similarly, in magnetic media, the magnetization results from the alignment of microscopic magnetic moments (originating from electron orbital motion and spin). Experimentally, the response of these moments is most naturally described in terms of the auxiliary field $\mathbf{H}$, which represents the externally applied magnetic field excluding the contribution of the material's own magnetization. If one attempted to write $\mathbf{M}$ as a function of $\mathbf{B}$ instead:

$$\mathbf{M} = \frac{1}{\mu_0}\chi'_m \mathbf{B} = \chi'_m(\mathbf{H} + \mathbf{M})$$

the relation would become implicit because $\mathbf{B}$ already contains the magnetization. But sure, you can write down $\mathbf{M} = (1/\mu_0)\chi' \mathbf{B}$ legally, it will becomes

$$\mathbf{M} = \frac{\chi'_m}{1 - \chi'_m}\mathbf{H} \Rightarrow \frac{\chi'_m}{1 - \chi'_m} \equiv \chi_m.$$

This is not convenient for experiments, just like you write down:

$$\mathbf{P} = \chi'_e \mathbf{D} = \chi'_e(\epsilon_o \mathbf{E} + \mathbf{P}) \Rightarrow \mathbf{P} = \epsilon_0 \frac{\chi'_e}{1 - \chi'_e}\mathbf{E} \Rightarrow \frac{\chi'_e}{1 - \chi'_e} \equiv \chi_e$$

Then the relations between (1) $\mathbf{D}$ and $\mathbf{E}$; (2) $\mathbf{H}$ and $\mathbf{B}$ become:

$$\begin{cases} \mathbf{D} = \epsilon_0 \dfrac{1}{1 - \chi'_e}\mathbf{E} \\ \mathbf{B} = \mu_0 \dfrac{1}{1 - \chi'_m}\mathbf{H}. \end{cases}$$

This symmetry is primarily algebraic rather than physical. In matter, the microscopic response of charges and magnetic moments is directly driven by the local electric and magnetic fields. Electric polarization arises from the displacement of bound charges under the action of the electric field **E**, while magnetization originates from the alignment of microscopic magnetic moments in the presence of the magnetic field **H**. By contrast, **D** and **B** already include contributions from the material response itself through **P** and **M**. Expressing the constitutive relations in terms of **D** or **B** therefore obscures the physical cause–response relationship and introduces unnecessary implicit dependence. Consequently, although the rewritten relations exhibit a formally symmetric structure, they are generally less intuitive and less convenient for describing the physical response of linear media.
Similarly, the auxiliary field is not always parallel to **B** since

$$\boldsymbol{\nabla} \cdot \mathbf{H} = -\boldsymbol{\nabla} \cdot \mathbf{M}, \tag{27}$$

where $\boldsymbol{\nabla} \cdot \mathbf{B} = 0$. Only the magnetic dipole moment is divergenceless, $\boldsymbol{\nabla} \cdot \mathbf{H} = \boldsymbol{\nabla} \cdot \mathbf{B} = 0$ occurs. It does not always happen.

A Brief Comparison

The laws modified by the fields in the matters:

$$\text{Gauss's Law}\begin{cases}\boldsymbol{\nabla} \cdot \mathbf{D} = \varrho_f \\ \displaystyle\int \mathbf{D} \cdot d\mathbf{a} = Q_{f_{\text{enc}}}\end{cases} \qquad \text{Ampère's law}\begin{cases}\boldsymbol{\nabla} \times \mathbf{H} = \mathbf{J}_f \\ \displaystyle\oint \mathbf{H} \cdot d\boldsymbol{\ell} = I_{f_{\text{enc}}}\end{cases}$$

Appendix: The Capacitor with a Linear Dielectric in Griffith

There's a configuration of the capacitor with voltage V, and the relation between the electric field and the voltage is:

$$-\int \mathbf{E} \cdot d\boldsymbol{\ell} = Ed = V \Rightarrow E = \frac{V}{d}, \tag{28}$$

as shown below

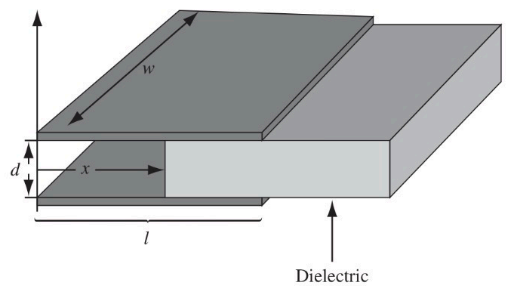

Figure 1: The Configuration of a Capacitor with a Linear Dielectric with Given Potential V

There are two regions: one is the vacuum region, and the other is the dielectric. The boundary condition of the infinite plate is given by

$$\mathbf{E} = \frac{\sigma}{\epsilon_0}\hat{\mathbf{n}}, \qquad E = \frac{\sigma}{\epsilon_0}.$$

So the charge density on the plate of the vacuum side is:

$$\sigma = \epsilon_0 E = \underbrace{\frac{\epsilon_0 V}{d}}_{\text{By (28)}}. \tag{29}$$

While the dielectric side is special because the dielectric is polarized by the infinite plane, this induced the bound charge density σ_b on the top of the dielectric, and it is

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = -\frac{\epsilon_0 \chi_e V}{d}, \tag{30}$$

where the minus sign means that the charge distribution has the opposite polarity to that on the plates. For example, if the upper plate carries positive charges, the dielectric becomes polarized, resulting in negative bound charges appearing on its upper surface. The dielectric becomes polarized and produces bound charges of opposite sign on its surfaces. These bound charges partially cancel the electric field produced by the free charges on the plates. Therefore, in order to maintain the same electric field between the plates, more free charge σ_f must accumulate on the plates in the region containing the dielectric. The free charge density is denoted by:

$$\sigma = \sigma_f + \sigma_b \Rightarrow \frac{\epsilon_0 V}{d} = \sigma_f - \frac{\epsilon_0 \chi_e V}{d},$$

and it leads to

$$\sigma_f = \frac{\epsilon_0(1+\chi_e)V}{d}. \tag{31}$$

It's obvious that $\sigma_f > \sigma$, which verifies the long-winded discussion above. So, the total charge Q_{tot} of the system is

$$\begin{aligned} Q_{\text{tot}} &= A_{\text{vac}}\sigma + A_{\text{die}}\sigma_f \\ &= wx\left(\frac{\epsilon_0 V}{d}\right) + w(l-x)\left[\frac{\epsilon_0(1+\chi_e)V}{d}\right] \\ &= \frac{w\epsilon_0 V}{d}\Big[x + (l-x)(1+\chi_e)\Big] = \frac{w\epsilon_0 V}{d}\Big[l\underbrace{(1-\chi_e)}_{\epsilon_r} - \chi_e x\Big] \\ &= \frac{w\epsilon_0 V}{d}\Big(\epsilon_r l - \chi_e x\Big). \end{aligned} \tag{32}$$

By the definition of capacitance $C = Q_{\text{tot}}/V$, we can substitute (32) into it. Therefore, we obtain the capacitance of this capacitor under this specific configuration:

$$\boxed{C = \frac{w\epsilon_0}{d}\Big(\epsilon_r l - \chi_e x\Big).} \tag{33}$$

It wasn't explained in Griffith, so ... You're welcome!

Some Identities

Triple Products:

$$\begin{cases} \mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \mathbf{B}\cdot(\mathbf{C}\times\mathbf{A}) = \mathbf{C}\cdot(\mathbf{A}\times\mathbf{B}) \\ \mathbf{A}\times(\mathbf{B}\times\mathbf{C}) = \mathbf{B}(\mathbf{A}\cdot\mathbf{C}) - \mathbf{C}(\mathbf{A}\cdot\mathbf{B}) \quad \text{Bac(k)-Cab law} \end{cases}$$

Products Rules:

$$\begin{cases} \boldsymbol{\nabla}(fg) = f(\boldsymbol{\nabla}g) + g(\boldsymbol{\nabla}f) \\ \boldsymbol{\nabla}(\mathbf{A}\cdot\mathbf{B}) = \mathbf{A}\times(\boldsymbol{\nabla}\times\mathbf{B}) + \mathbf{B}\times(\boldsymbol{\nabla}\times\mathbf{A}) + (\mathbf{A}\cdot\boldsymbol{\nabla})\mathbf{B} + (\mathbf{B}\cdot\boldsymbol{\nabla})\mathbf{A} \\ \boldsymbol{\nabla}\cdot(\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot(\boldsymbol{\nabla}\times\mathbf{A}) - \mathbf{A}\cdot(\boldsymbol{\nabla}\times\mathbf{B}) \\ \boldsymbol{\nabla}\cdot(f\mathbf{A}) = f(\boldsymbol{\nabla}\cdot\mathbf{A}) + \mathbf{A}\cdot\boldsymbol{\nabla}f \\ \boldsymbol{\nabla}\times(f\mathbf{A}) = f(\boldsymbol{\nabla}\times\mathbf{A}) - \mathbf{A}\times\boldsymbol{\nabla}f \\ \boldsymbol{\nabla}\times(\mathbf{A}\times\mathbf{B}) = (\mathbf{B}\cdot\boldsymbol{\nabla})\mathbf{A} - (\mathbf{A}\cdot\boldsymbol{\nabla})\mathbf{B} + \mathbf{A}(\boldsymbol{\nabla}\cdot\mathbf{B}) - \mathbf{B}(\boldsymbol{\nabla}\cdot\mathbf{A}) \end{cases}$$

Second Derivatives:

$$\begin{cases} \boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\mathbf{A}) = 0 \\ \boldsymbol{\nabla}\times(\boldsymbol{\nabla}f) = \mathbf{0} \\ \boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\mathbf{A}) = \boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\mathbf{A}) - \nabla^2\mathbf{A} \end{cases}$$

Green's Identities and Some Integrals:

$$
\begin{cases}
\displaystyle\int_{\mathcal{V}} \left(T\,\nabla^2 U - U\nabla^2 T\right) d\tau = \oint_{\mathcal{S}} \left(T\boldsymbol{\nabla} U - U\boldsymbol{\nabla} T\right)\cdot d\mathbf{a} \\
\displaystyle\int_{\mathcal{S}} \boldsymbol{\nabla} T \times d\mathbf{a} = -\oint_{\mathcal{P}} T\,d\boldsymbol{\ell} \\
\displaystyle\oint (\mathbf{c}\cdot\mathbf{r})\,d\boldsymbol{\ell} = \left(\int d\mathbf{a}\right)\times\mathbf{c} \\
\displaystyle\int_{\mathcal{V}} \boldsymbol{\nabla}\times\mathbf{F}\,d\tau = -\oint_{\mathcal{S}} \mathbf{F}\times d\mathbf{a}
\end{cases}
$$

Delta functions:

$$
\begin{cases}
\boldsymbol{\nabla}\cdot\left(\dfrac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}\right) = 4\pi\delta^3(\boldsymbol{\mathscr{r}}) \\
\boldsymbol{\nabla}\left(\dfrac{1}{\mathscr{r}}\right) = \boldsymbol{\nabla}\left(\dfrac{1}{\mathbf{r}-\mathbf{r}'}\right) = -\dfrac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2} \\
\boldsymbol{\nabla}'\left(\dfrac{1}{\mathscr{r}}\right) = \boldsymbol{\nabla}'\left(\dfrac{1}{\mathbf{r}-\mathbf{r}'}\right) = \dfrac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2} \\
\nabla^2\left(\dfrac{1}{\mathscr{r}}\right) = -4\pi\delta^3(\boldsymbol{\mathscr{r}})
\end{cases}
$$